

Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

B.Tech Degree S5 (R, S) / S5 (PT) (R, S) / S3 (PT) (R, S) Examination December 2023 (2019 Scheme)

**Course Code: HUT 310****Course Name: MANAGEMENT FOR ENGINEERS**

Max. Marks: 100

Duration: 3 Hours

**PART A***(Answer all questions; each question carries 3 marks)*

Marks

- |    |                                                                                                                             |   |
|----|-----------------------------------------------------------------------------------------------------------------------------|---|
| 1  | “Effectiveness and efficiency balance is a main feature of management.” Summarize.                                          | 3 |
| 2  | Discuss the aspects in which the neoclassical theory improved the classical theory.                                         | 3 |
| 3  | Enumerate the advantages of functional organisation?                                                                        | 3 |
| 4  | Explain span of control.                                                                                                    | 3 |
| 5  | “A firm gain competitive advantage through differentiation of their products”. Elaborate                                    | 3 |
| 6  | Describe the economic man model.                                                                                            | 3 |
| 7  | Identify the methodology which is opted when activity timings could not be estimated with enough certainty? List the steps. | 3 |
| 8  | Describe the different start and finish times of an activity.                                                               | 3 |
| 9  | Discuss the four P’s of marketing mix.                                                                                      | 3 |
| 10 | Explain Cash Budget.                                                                                                        | 3 |

**PART B***(Answer one full question from each module, each question carries 14 marks)***Module -1**

- |    |                                                                                                                     |   |
|----|---------------------------------------------------------------------------------------------------------------------|---|
| 11 | a) State and explain the managerial roles as explained by Henry Mintzberg. Clearly identify how each is subdivided. | 8 |
|    | b) Do managers at all levels require managerial skills? Substantiate with suitable examples.                        | 6 |
| 12 | a) Explain how the managerial functions interrelates with each other                                                | 8 |
|    | b) “Management is a composite and continuous process.” Express your views.                                          | 6 |

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### Module -2

- 13 a) List the features of management process? 6  
b) Define the terms, Authority, Responsibility and Accountability. Also, list the sources of authority? 8
- 14 a) Describe Objectives and classify the same. 6  
b) Suppose the chairman of XYZ company is planning to divide the authority and work through a systematic division of labor. Describe the factors that have to be taken into account while making final decision? 8

### Module -3

- 15 Explain in detail different productivity measurement indexes 14
- 16 a) A food products company is planning the introduction of a revolutionary new product with new packing to replace the existing product at much higher price (S1) or a moderate change in the composition of the existing product with a new packaging at a small increase in price (S2) or a small change in the composition of the existing except the word, 'New' with a negligible increase in the price (S3). The three possible states of nature of events are (i) high increase in sales (N1) (ii) no change in sales (N2) (iii) decrease in sales (N3). The marketing department of the company worked out the payoffs in terms of yearly new profits for each of the strategies on these events. This is represented in the following table.  
Which strategy should the executive concerned choose on the basis of  
(a) Maximin Criterion  
(b) Maximax Criterion  
(c) Minimax regret Criterion  
(d) Laplace criterion

|            | Pay offs         |     |     |
|------------|------------------|-----|-----|
|            | States of nature |     |     |
| Strategies | N1               | N2  | N3  |
| S1         | 700              | 300 | 150 |
| S2         | 500              | 450 | 0   |
| S3         | 300              | 300 | 300 |

- b) Explain programmed and non programmed decisions 4

### Module -4

- 17 a) Explain Activity-on-Arc(AOA) and Activity-on-Node(AON) in detail with suitable graphical representations 8

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- b) Discuss the time-cost optimization procedure in detail 6
- 18 a) For the given activities determine: 10
1. Critical path using PERT
  2. Calculate variance and standard deviation for each activity.
  3. Calculate the probability of completing the project in 26 days.

| Activity | $t_o$ | $t_m$ | $t_p$ |
|----------|-------|-------|-------|
| 1-2      | 6     | 9     | 12    |
| 1-3      | 3     | 4     | 11    |
| 2-4      | 2     | 5     | 14    |
| 3-4      | 4     | 6     | 8     |
| 3-5      | 1     | 1.5   | 5     |
| 2-6      | 5     | 6     | 7     |
| 4-6      | 7     | 8     | 15    |
| 5-6      | 1     | 2     | 3     |

- b) What are the different networking components 4

### Module -5

- 19 a) “Financial Management is managing the finances through scientific decision making”. Defend stating the different financial management functions. 10
- b) Explain Holistic Marketing concept 4
- 20 a) “While acquiring new employees into a firm, training is needed for their adaptation”. Discuss. 8
- b) How an entrepreneurship is classified based on its leadership? Explain in detail. 6

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